

SHAHRIAR RATUL

Software Engineer

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SUMMARY

Passionate about leveraging technology and data-driven insights to improve processes and solve real-world problems across technology and business strategy. I am skilled at collaborating with diverse team and bridging the gap between technical insights and business outcomes.

SKILLS

Programming

Python, C++, C, Java, JavaScript, TypeScript, Dart

Data & AI

PyTorch, NumPy, Pandas, Matplotlib, Power BI, Feature Engineering

Full-Stack

FastAPI, React, Node.js, Next.js, Flutter, MongoDB, PostgreSQL, Firebase

Robotics & Design

ESP32, Arduino, Raspberry Pi, LQR/PID, IK, OnShape, 3D Printing

Specializations

Computer Vision, Object Detection, Knowledge Distillation, IoT Architecture

Tools

Git, GitHub, Docker, Kubernetes, Linux, VS Code, MS Excel

Soft Skills

Team Leadership, Communication, Conflict Resolution

EXPERIENCE

Intern – AI Engineer

Cloudly Infotech

02/2026 – Present 📍 Dhaka, Bangladesh

- Currently contributing on CloudlyIO's RADP (RAN Intelligent Controller Algorithm Development Platform) project "NetAI". NetAI is a multi-tenant SaaS platform that leverages Bayesian Digital Twins and multi-agent Reinforcement Learning to autonomously optimize cellular networks for energy efficiency, coverage, and throughput.
- Research and academic paper writing titled "Digital-Twin Assisted Reinforcement Learning for Mobility Robustness Optimization in 5G RANs". The work leverages Bayesian Digital Twin and Proximal Policy Optimization (PPO) (RL) to adjust cellular handover parameters HYST and TTT.
- Learned the complete software development lifecycle hands-on. Worked on live projects with professional teams. Adopted industry-standard practices and development tools like docker, redis, kafka etc.

EDUCATION

BSc in Computer Science and Engineering North South University

06/2021 – 09/2025 📍 Dhaka, Bangladesh
CGPA: 3.01

Higher Secondary Certificate Shaheed Police Smriti College

2019 📍 Dhaka, Bangladesh
GPA: 4.75

Secondary School Certificate Shaheed Police Smriti College

2017 📍 Dhaka, Bangladesh
GPA: 5.00

EXTRA-CURRICULAR

Captain – NSU Volleyball Team

Started from school and continued this passion till university. Participated in various tournaments on behalf of NSU. Had the honor to lead as the captain of the NSU volleyball team.

NSU Games & Sports Club

Organized events, friendly matches and tournaments, Conducted sports tryouts for volleyball, and managed operations.

NSU Intelligence & Robotics Labs (NIRO)

Was very active in NIRO Labs since its birth in 2025. Experimented with 3D printing, hands-on programming on microcontrollers, and conducted workshops on robotics and embedded systems programming. Participated in the first-ever BEAR Summit / National Semiconductor Symposium, Bangladesh– showcasing robotics projects in a national exhibition, demonstrating the innovation and technical expertise of NIRO Labs.

PROJECTS

NSU Career and Placement Center Website

 Repo/  Video

2023

- Replaced the legacy PHP/HTML university job portal with a modern MERN stack, significantly improving UI/UX and streamlining the job placement workflow.
- Integrated Google Authentication for secure single-sign-on login and built an automatic CV generator that populates from student profile data.

Stack: MongoDB, Express, React, Node.js, Tailwind CSS

Knowledge Distillation on YOLO with KAN

 Drive

2024

- Implemented and analyzed YOLO8x, YOLO5x, YOLO8n, and YOLO5n with Kolmogorov-Arnold Networks using knowledge distillation across COCO, Crack-seg, Braintumor, and African-wildlife datasets.
- Produced smaller, more efficient student models without sacrificing detection accuracy – validating KAN as a promising distillation target architecture.

Stack: Python, PyTorch, OpenCV, Kolmogorov-Arnold Networks

ACROBOT – Robots With Inverted Pendulums

 Repo

2025

- Designed and fabricated a hybrid hexapod walking robot and a cube robot with an inverted-pendulum active balancing system that balances on its edges.
- Implemented LQR controller in MATLAB, Madgwick filter for IMU fusion, and Inverse Kinematics for gait planning; simulated in PyBullet before hardware deployment.

Stack: MATLAB, ESP32, Arduino Uno, MPU6050, LQR/PID, PyBullet, OnShape, 3D Printing

SENTINEL – A Home Security System

 Repo  Demo

2025

- Built a multi-sensor smart home security system covering sonar, IR, temperature, humidity, smoke, gas leak detection, and a password-protected door lock.
- Developed the companion Flutter app “SENTINEL” with a Firebase backend for real-time sensor dashboards, push-based intruder alerts, remote monitoring, and Bluetooth voice control for fan/light automation.

Stack: Flutter, Dart, Python, Raspberry Pi Pico W, Firebase

Scout – Volleyball Players Portfolio Platform And Live Referee App

 Repo  Referee App

2026

- Built a full-stack platform for the Bangladeshi volleyball community where players sign in, create stat portfolios, and get discovered by teams and coaches.
- Developed a companion VB Referee app for conducting live matches and tracking scores in real time.

Stack: FastAPI, Next.js, MongoDB, Docker, GitHub Actions

Plant Disease Detection and Quality Classification

 Repo

2024

- Built a custom CNN classifying 38 plant-disease classes from 128×128 leaf images, demonstrating end-to-end deep learning model design for agricultural applications.
- Trained and evaluated model performance across multiple class-imbalanced categories using cuDNN-accelerated training.

Stack: Python, PyTorch, cuDNN, Kaggle Plant Disease Dataset

REFERENCES

Dr. Shahnewaz Siddique

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Dept. of Electrical & Computer Engineering

North South University, Dhaka

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